

Stepwise Multi-Objective Finetuning for Triple-Horizon Anomaly Detection: Horizon-Specific Loss Extension via Pareto-Guided Darwin-Gödel Machine

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Abstract

Anomaly detection in pump-facility time series is hindered by severe class imbalance—fault events occur in fewer than 13% of observation windows—and by the dependency of the optimal loss configuration on the forecast horizon. Jointly optimizing all hyperparameters (architecture, adapter rank, loss function, and class weights) causes cross-subspace interference that degrades Pareto-frontier convergence and wastes the evaluation budget.

We propose **Stepwise Multi-Objective DGM**, a framework that applies the Darwin-Gödel Machine (DGM) self-improvement loop in four sequential phases, each targeting a distinct parameter subspace while freezing prior discoveries. Phase 1 selects PatchTST via comparative LoRA fine-tuning (AUC = 0.965 vs. 0.529–0.770 for three alternatives). Phase 2 fixes LoRA rank = 16 via systematic rank ablation. Phase 3 optimizes Focal Loss and class weights over a 4D space via NSGA-II DGM (macro-F1 = 0.776, 500 iterations, 89.6% Pareto rate). Phase 4 optimizes PatchTST patch granularity, establishing patch.len = 26, stride = 16 as the reproducible optimum (macro-F1 = 0.773). However, the stepwise approach has a limitation in that it may not fully capture interactions between parameter subspaces, and fixed hyperparameter values from a prior phase narrow the scope for post-phase search improvement. Simultaneous optimization of all hyperparameter subspaces could potentially address these limitations, but it introduces cross-subspace interference that complicates Pareto-frontier convergence and increases the evaluation budget.

Building on these fixed values, we further propose **Horizon-Specific 9D DGM**: each forecast horizon $h \in \{30, 60, 90\}$ receives independent Focal Loss parameters $(\alpha_h, \gamma_h, w_{n,h})$, subject to the per-horizon normalisation constraint $w_{n,h} + w_{a,h} = W_{\text{sum}}$. This design decouples the inherently different loss landscapes of short-, medium-, and long-range prediction. Systematic 4D→6D→8D ablation studies con-

firm that the phased approach outperforms full joint search by 12.0 points in macro-F1 (0.776 vs. 0.656), validating the structural decomposition. A transferable finding also emerges: $\alpha_L/r \approx 2.9$ is a stable LoRA scaling law independent of search dimensionality. Under this stable LoRA condition, the focal loss and class-weight parameter subspace can be effectively optimized to improve horizon-specific anomaly detection performance. This horizon-specific loss design enables tailored 3D optimization under the class-weight-sum constraint for each of the triple forecast horizons, enhancing detection accuracy across short-, medium-, and long-range predictions.

Keywords: Anomaly Detection, PatchTST, Darwin-Gödel Machine, Multi-Objective Pareto Optimization, NSGA-II, Horizon-Specific Focal Loss.

1 Introduction

1.1 Background and Motivation

Infrastructure facilities such as pump equipment require continuous condition monitoring to prevent unexpected failures [1]. Time-series anomaly detection using transformer architectures has shown promise, yet the class imbalance between normal and anomalous states—fault events occur in fewer than 13% of time windows—remains a fundamental challenge.

PatchTST [2] is a patch-based time-series transformer that treats temporal patches as input tokens, analogous to ViT [3] in vision. When fine-tuned with LoRA [4] for anomaly classification, it achieves competitive detection metrics on pump-facility sensor data. Focal Loss [5] addresses class imbalance by down-weighting easy examples via focusing factor γ , combined with per-sample class weights. However, the combined hyperparameter space—model architecture, adapter rank, and loss function—spans many interdependent dimensions and resists analytical treatment.

1.2 Darwin-Gödel Machine for Horizon-Specific Improvement

The Darwin-Gödel Machine (DGM) [6] operationalises open-ended self-improvement: LLM-based coding agents propose parameter modifications, a fitness function evaluates each proposal, and accepted solutions seed future generations. In a preceding study [7], we applied DGM to 10-class damage-cause encoding, demonstrating that NSGA-II multi-objective DGM yields 11 Pareto-optimal solutions (macro-accuracy 0.840–0.958) from a single 50-iteration run. This paper extends DGM with a *phased strategy*: each run targets one parameter subspace at a time. The stepwise approach mitigates cross-subspace interference by isolating the optimization of each hyperparameter subspace under prior phase discoveries. Furthermore, this phased strategy is naturally extended to simultaneous hyperparameter search: a 4D, 6D, 8D gradual progression for effective range exploration in each horizon-consistent subspace. However, simultaneous search across multiple subspaces has limitations in improving horizon-specific accuracy due to the increased complexity of horizon feature representation. This paper proposes a stepwise, horizon-specific optimization framework for hyperparameter search, mitigating cross-subspace interference and enhancing triple-horizon improvement.

1.3 Contributions

- Stepwise Multi-Objective DGM (4 Phases):** Principled decomposition of the PatchTST hyperparameter space; each phase freezes prior discoveries to prevent cross-subspace interference. Model Selection (Phase 1): Comparative LoRA finetuning of AnomalyBERT, PatchTST, Chronos [8], and TimesFM [9]; PatchTST selected for AUC = 0.965(60d). LoRA Ablation (Phase 2): Systematic study of rank = 16. 4D Focal Loss DGM (Phase 3): Best: $\alpha=0.866$, $\gamma=1.156$, $w_n=1.851$, $w_a=4.035$. 2D Architecture DGM (Phase 4): optimum at $L_p=26$, $s=16$.
- Structured Joint optimization:** Systematic 4D→6D→8D progression (F1: 0.614→0.642→0.656); three empirical findings—LoRA $\alpha/r \approx 2.9$ scaling law, near-balanced class weights optimal under Focal Loss, and weak cross-component interactions—confirm the stepwise decomposition as the correct structural prior.
- Horizon-Specific 9D DGM:** Extends the joint optimization framework to assign independent Focal Loss parameters ($\alpha_h, \gamma_h, w_{n,h}$) per forecast horizon $h \in \{30, 60, 90\}$ d, subject to $w_{n,h} + w_{a,h} =$

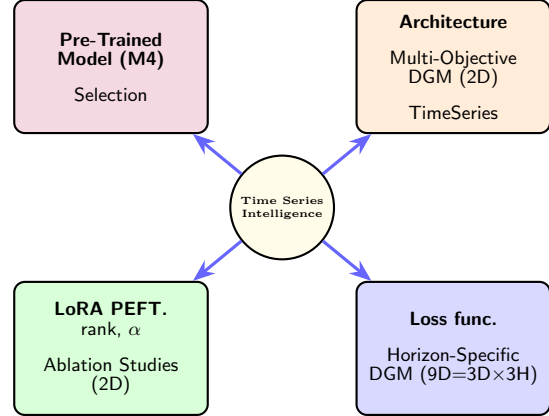


Figure 1: Phased DGM framework for PatchTST optimization. The central AI orchestrates four specialised phases: (1) Model Selection from pre-trained M4 candidates, (2) LoRA PEFT Ablation Studies and Multi-Objective DGM (2D: rank, scaling), (3) Horizon-Specific Class Weighted Focal Loss DGM (9D=3D×3H, constraint on the sum of weights), (4) Architecture DGM (2D: patch.len, stride). Each phase freezes prior discoveries to prevent cross-subspace interference.

5.88. Extended to 3,000 iterations, the search attains a best macro-F1 of 0.769 (30d F1 = 0.789), surpassing the 8D joint baseline by +11.3 points.

1.4 Phased DGM Framework and Horizon-Specific Extension

Figure 1 illustrates the four-phase intelligence harness framework. Each phase targets a distinct parameter subspace, with outputs feeding forward to constrain subsequent phases. The central AI-driven DGM loop orchestrates multi-objective optimization across time-series model selection, LoRA ablation, loss function tuning, and architecture constraint enforcement. In the stepwise multi-objective DGM framework, each phase incrementally refines a horizon consistent subset of hyperparameters: architecture (patch length, stride) and parameter efficient fine-tuning (LoRA r, α). However, the loss function parameters (focal loss α, γ and class weights w_n, w_a) are optimized horizon-specifically to account for their distinct effects on the three forecast horizons’ performance. Therefore, this paper extends to a horizon-specific 9D (=3D×3H) DGM framework that separates the loss-function hyperparameters across different forecast horizons under the constraint on the sum of class weights.

2 Related Work

2.1 Time-Series Transformer Architectures for Anomaly Detection

PatchTST [2] achieves strong long-term forecasting performance by segmenting the input into patch tokens, analogous to ViT [3] in computer vision. The patch-based tokenization reduces sequence length and improves local temporal coherence. The original Transformer [10] established the self-attention mechanism underlying all modern sequence models; Informer [11] reduced its quadratic complexity to $O(N \log N)$ via ProbSparse attention, enabling long-range forecasting. AnomalyBERT [12] applies BERT-style self-supervised pre-training to time-series anomaly detection. Foundation models Chronos [8] (T5-based) and TimesFM [9] (decoder-only) offer zero-shot generalization but require substantially more parameters for effective fine-tuning on small domain-specific datasets. Our Phase 1 comparison demonstrates that PatchTST achieves superior AUC/FPR balance on pump-facility data with only 49K trainable parameters, at a fraction of the cost of foundation models.

Transformer-based anomaly detectors have advanced the state of the art on multivariate sensor data. Anomaly Transformer [13] introduces an association-discrepancy criterion that measures the divergence between prior association and series association to identify anomalies without requiring labels. OmniAnomaly [14] uses stochastic recurrent networks (GRU-VAE) with planar normalizing flows to model the distribution of normal multivariate time series. USAD [15] employs adversarially trained autoencoders to amplify reconstruction errors for anomalous inputs. TranAD [16] combines a Transformer encoder with adversarial training and a focus score mechanism tailored for multi-entity industrial IoT data. In contrast to these unsupervised reconstruction-based methods, our approach operates in a supervised binary-classification setting with labelled inspection records, enabling direct optimization of precision, recall, and FPR per forecast horizon.

2.2 Focal Loss, Class-Imbalance Learning and PEFT

Focal Loss [5] was originally proposed for dense object detection, down-weighting easy examples via the focusing factor γ . For binary anomaly detection, class imbalance is severe: fault events may occur in fewer than 10% of time windows. Standard cross-entropy

biases models toward the majority class; Focal Loss mitigates this but requires careful joint tuning of α , γ , and per-class weights. SMOTE [17] addresses imbalance by synthesizing minority-class examples through feature-space interpolation, but cannot be straightforwardly applied to sequential sensor data without disrupting temporal structure. Cost-sensitive learning [18] adjusts the misclassification penalty asymmetrically, a strategy that overlaps with per-class weight tuning. The present paper automates this joint tuning via multi-objective DGM, removing the need for manual heuristic selection.

LoRA [4] inserts low-rank adapter matrices into frozen Transformer weight matrices, enabling efficient fine-tuning with approximately 1–10% of total parameters. The rank r and scaling factor α_L jointly control the adapter capacity; the ratio α_L/r governs the effective learning rate of the adapter. Phase 2 ablation over $r \in \{8, 12, 16, 20, 32, 64\}$ confirms rank = 16 as the Pareto-optimal operating point for PatchTST on the pump-facility Golden Testset, consistent with findings from NAS [19] that moderate-capacity adapters outperform both under- and over-parameterized configurations.

2.3 Multi-Objective Evolutionary Computation and Hyperparameter Optimization

NSGA-II [20] remains the standard baseline for multi-objective evolutionary optimization due to its efficient non-dominated sorting and crowding-distance diversity mechanism. NSGA-III [21] extends NSGA-II to many-objective problems (≥ 4 objectives) via reference-point-based selection, and is better suited to objectives exceeding eight dimensions. SPEA2 [22] maintains an external archive of non-dominated solutions and uses a fitness assignment strategy based on strength values and raw fitness. The hypervolume indicator [23] provides a unary quality measure of a Pareto approximation set; its computation is exponential in the number of objectives, but feasible for the 15-objective vector used in this work. Our framework wraps NSGA-II via Optuna [24] with a tell/ask interface that enables the LLM validator to intercept and micro-correct proposals before evaluation.

Hyperband [25] achieves near-optimal simple regret by adaptively allocating the training budget among configurations using successive halving. BOHB [26] combines Hyperband with Bayesian optimization (TPE [27]) to exploit model-based guidance while retaining Hyperband’s early-stopping efficiency. Neural Architecture Search (NAS) [19] automates model design via evolutionary, gradient-based, or

reinforcement-learning strategies; our Phase 4 maps to a small 2D grid-constrained NAS over patch granularity. FunSearch [28] uses LLMs to evolve mathematical programs that improve on human-designed baselines, demonstrating the generality of LLM-guided search beyond software engineering tasks. The DGM framework [6] and our extension differ from AutoML pipelines in that the LLM validates proposals with domain-specific anomaly detection context rather than treating the search as a black-box function evaluation.

2.4 Self-Improvement Agents and Darwin-Gödel Machine

LLM-based self-improvement. Early work on self-improving agents focused on verbal reinforcement: Reflexion [29] stores episodic feedback in a linguistic memory and revisits failed trajectories without weight updates; ReAct [30] interleaves reasoning traces with environment actions, enabling chain-of-thought grounding. ExpEL [31] extends this with experiential replay, accumulating task-specific insight across episodes. PromptBreeder [32] automates meta-prompt evolution via self-referential mutation, showing that the instruction itself can be the optimisation target.

Open-ended and automated discovery. OPRO [33] frames optimisation as LLM meta-prompting, solving combinatorial tasks by iteratively refining solution candidates in context. ADAS [34] searches the space of agentic system designs, evolving agent architectures rather than fixed hyperparameters. AI Scientist [35] automates end-to-end research: idea generation, experiment execution, result analysis, and paper writing. Quiet-STaR [36] teaches models to produce internal rationales before committing to outputs, improving sample efficiency in few-shot settings. LLM-guided EMO [37] integrates LLM proposals directly into evolutionary multi-objective optimisation loops, reducing function evaluations needed to approximate the Pareto front. Agentless [38] demonstrates that structured, plan-then-execute agents can match or exceed complex agent frameworks on software engineering benchmarks at lower cost.

Darwin-Gödel Machine. DGM [6] operationalizes the Gödel Machine [39] framework with LLM-based coding agents validated on SWE-bench. Eureka [40] demonstrates LLM-designed reward functions that surpass human experts on robotic control. In a preceding study [7], we applied DGM to 10-class damage-cause encoding, demonstrating that NSGA-II multi-objective DGM yields 11 Pareto-optimal solu-

tions (macro-accuracy 0.840–0.958) from a single 50-iteration run. The present paper scales DGM to a 15-objective, multi-phase hyperparameter search with a horizon-specific parameter design that is novel to the anomaly detection literature.

3 Problem Formulation

3.1 PatchTST Anomaly Classifier

Given a univariate time series $\mathbf{x} \in \mathbb{R}^T$ ($T=90$ days), PatchTST divides it into N_p patches of length L_p with stride s :

$$N_p = \left\lfloor \frac{T - L_p}{s} \right\rfloor + 1. \quad (1)$$

The patches are linearly projected and passed through a Transformer encoder [10] to produce classification logits $\hat{\mathbf{y}} \in \mathbb{R}^3$ for three binary anomaly horizons $h \in \mathcal{H} = \{30, 60, 90\}$ d. LoRA adapters (rank r , scaling α_L) are inserted into the attention projection layers; all backbone weights are frozen.

3.2 Horizon-Specific Weighted Focal Loss

For each horizon h and training sample i , the per-instance weighted Focal Loss is:

$$\ell_{i,h}(\boldsymbol{\theta}_h) = c_{i,h} \cdot \text{FL}(\hat{y}_{i,h}, y_{i,h}; \alpha_h, \gamma_h) \quad (2)$$

where the Focal Loss kernel [5] is

$$\text{FL}(p_t; \alpha_h, \gamma_h) = -\alpha_{y,h}(1 - p_t)^{\gamma_h} \log p_t, \quad (3)$$

$$\alpha_{y,h} = \begin{cases} \alpha_h & y=1, \\ 1 - \alpha_h & y=0, \end{cases} \quad (4)$$

$p_t = \sigma(\hat{y}_{i,h}) \cdot y_{i,h} + (1 - \sigma(\hat{y}_{i,h})) \cdot (1 - y_{i,h})$ is the model probability for the ground-truth class, and the per-sample class weight is

$$c_{i,h} = w_{a,h} y_{i,h} + w_{n,h} (1 - y_{i,h}). \quad (5)$$

The horizon-average training loss over N samples is:

$$\mathcal{L}(\boldsymbol{\Theta}_H) = \frac{1}{|\mathcal{H}|} \sum_{h \in \mathcal{H}} \frac{1}{N} \sum_{i=1}^N \ell_{i,h}(\boldsymbol{\theta}_h) \quad (6)$$

where $\boldsymbol{\Theta}_H = \{\boldsymbol{\theta}_h\}_{h \in \mathcal{H}}$ collects all horizon-specific parameters.

Threshold optimization. For each horizon h , the decision threshold is calibrated on the validation set by maximizing $F1$:

$$\tau_h^* = \arg \max_{\tau \in [0.1, 0.9]} F1(\{\hat{y}_{i,h}, y_{i,h}\}_{\text{val}}; \tau). \quad (7)$$

The threshold grid spacing is 0.01, giving 81 candidate values.

3.3 Multi-Objective Pareto Formulation

Objective vector. Each fine-tuning run with parameters θ produces a 15-dimensional evaluation vector:

$$\mathbf{f}(\theta) = \left(\underbrace{\text{AUC}_h, P_h, R_h, F1_h}_{\text{maximise, } \times 3}, \underbrace{-\text{FPR}_h}_{\text{maximise (negated), } \times 3} \right) \in \mathbb{R}^{15}. \quad (8)$$

FPR objectives are negated internally so that all 15 dimensions are maximised, enabling standard NSGA-II dominance comparison.

Pareto dominance. Solution θ *dominates* θ' , written $\theta \prec \theta'$, if and only if

$$f_j(\theta) \geq f_j(\theta') \quad \forall j \in [15] \quad \text{and} \quad \exists j : f_j(\theta) > f_j(\theta'). \quad (9)$$

The *Pareto front* is $\mathcal{P}^* = \{\theta \in \mathcal{H} \mid \nexists \theta' \in \mathcal{H} : \theta' \prec \theta\}$.

3.4 Phased Search Space Definitions

Phase 3 (4D Focal Loss). Architecture ($L_p=30$, $s=30$) and LoRA ($r=16$, $\alpha_L=32$) are fixed; the search space is:

$$\theta_3 = (\alpha_{\text{FL}}, \gamma, w_n, w_a), \quad \alpha_{\text{FL}} \in [0.1, 0.9], \quad \gamma \in [0.5, 5.0], \\ w_n \in [0.1, 5.0], \quad w_a \in [0.5, 10.0].$$

Phase 4 (2D Architecture). Focal Loss is fixed at Phase 3 optimal values; the search space is:

$$\theta_4 = (L_p, s), \quad L_p \in [20, 32], \quad s \in [10, 20].$$

Horizon-specific 9D (v0.2.1H). Architecture and LoRA are fixed; the search space is:

$$\Theta_H = \{(\alpha_h, \gamma_h, w_{n,h})\}_{h \in \mathcal{H}}, \quad (10)$$

$$\alpha_h \in [0.5, 0.9], \quad \gamma_h \in [0.6, 2.0], \quad w_{n,h} \in [0.3, 5.5], \quad (11)$$

with the per-horizon normalisation constraint $w_{n,h} + w_{a,h} = W_{\text{sum}} = 5.88$ for all h . The anomaly-class weight is derived as $w_{a,h} = W_{\text{sum}} - w_{n,h}$, reducing the free variables to three per horizon (nine total).

4 Methodology

4.1 Stepwise optimization Overview

Figure 2 illustrates the four-phase strategy. Each phase produces fixed optimal values passed forward as frozen hyperparameters to subsequent phases.

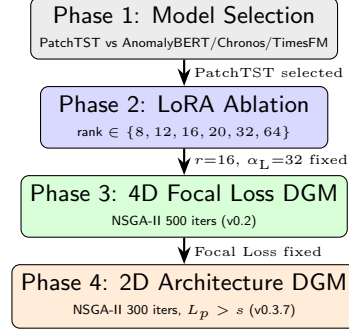


Figure 2: Four-phase stepwise optimization. Each phase freezes prior discoveries and passes optimal values to the next phase.

Table 1: Model selection (60d metrics, LoRA fine-tuned).

Model	Params	AUC	F1	FPR
PatchTST	49K	0.965	0.815	0.011
AnomalyBERT [12]	115K	0.770	0.378	0.139
Chronos [8]	1.18M	0.565	—	—
TimesFM [9]	2.46M	0.529	—	—

4.2 Phase 1: Model Selection

Four architectures were LoRA fine-tuned (rank 16) on pump and HVAC facility data and compared on 60d AUC (Table 1). PatchTST (AUC = 0.965, F1 = 0.815, FPR = 0.011) outperforms AnomalyBERT by +0.195 AUC and foundation models Chronos and TimesFM by large margins, using only 49K trainable parameters. PatchTST is selected for all subsequent phases.

4.3 Phase 2: LoRA Rank Ablation

We ablated rank $r \in \{8, 12, 16, 20, 32, 64\}$ with $\alpha_L = 2r$ on the Golden Testset (Table 2). Rank 8 shows degraded average AUC (0.745) due to insufficient adapter capacity. Ranks 20–64 add parameters without improving AUC over rank 16, indicating overfitting of the adapter. Rank 16 ($\alpha_L = 32$, 7.49% trainable parameters) is the Pareto-optimal operating point.

4.4 Phase 3: 4D Focal Loss DGM

With LoRA fixed at Phase 2 values, we applied the DGM loop to θ_3 for 500 iterations (v0.2). The DGM loop adopts the five-module design of [7]:

1. **Archive:** JSONL-backed store of all evaluated $(\theta, \mathbf{f})$ pairs.

Table 2: LoRA ablation: average AUC, F1, FPR across 30/60/90d.

r	%	AUC	F1	FPR
8	3.89	0.745	0.377	0.453
12	5.73	0.897	0.609	0.090
16	7.49	0.947	0.741	0.022
20	9.19	0.909	0.631	0.102
32	13.94	0.902	0.627	0.082
64	24.47	0.911	0.631	0.105

2. **NSGA-II Sampler:** Optuna NSGA-II (population = 20) proposes θ exploring the 15-objective Pareto front.
3. **LLM Validator (Codestral 22B):** Validates the NSGA-II proposal; applies micro-corrections if counter-productive; falls back to raw NSGA-II on failure.
4. **Evaluator:** Fine-tunes PatchTST (100 epochs, early stopping, patience = 20) and returns the 15-element objective vector.
5. **Pareto Archive:** Maintains the non-dominated front; FPR objectives internally negated for “higher-is-better” dominance comparison.

4.5 Structured Joint Optimization: 4D, 6D, and 8D

To characterize how joint optimization scales with dimensionality, we designed three systematic experiments (v0.2.1–v0.2.3) that progressively widen the search space while holding the DGM loop structure constant.

v0.2.1 (4D: Focal Loss and Class Weights). Architecture ($L_p=29$, $s=16$) and LoRA ($r=19$, $\alpha_L=53$) are fixed; the four parameters ($\alpha_{FL}, \gamma, w_n, w_a$) are jointly optimized via NSGA-II for 1000 iterations.

v0.2.2 (6D: Architecture + LoRA + Focal Loss). Class weights are fixed at v0.2.1 best ($w_n=1.717$, $w_a=2.463$); the six parameters ($L_p, s, r, \alpha_L, \alpha_{FL}, \gamma$) are jointly optimized for 2000 iterations.

v0.2.3 (8D: Full joint). All eight parameters are simultaneously optimized for 3000 iterations, covering Architecture, LoRA, Focal Loss, and Class Weights (Table 3). No parameter is fixed.

Table 3: Structured joint optimization configurations.

Parameter	v0.2.1 (4D)	v0.2.2 (6D)	v0.2.3 (8D)
L_p	29 (fixed)	[24–32]	[24–32]
s	16 (fixed)	[14–18]	[14–18]
r	19 (fixed)	[15–25]	[15–25]
α_L	53 (fixed)	[45–60]	[45–60]
α_{FL}	[0.5–0.9]	[0.7–0.9]	[0.7–0.9]
γ	[0.6–2.0]	[0.6–1.2]	[0.6–1.2]
w_n	[1.0–3.0]	1.717 (fixed)	[1.0–3.0]
w_a	[2.0–6.0]	2.463 (fixed)	[2.0–6.0]
Budget	1000	2000	3000
NSGA-II μ	20	20	20

4.6 Horizon-Specific 9D DGM

4.6.1 Motivation

The joint experiments (v0.2.1–v0.2.3) apply a *single* loss configuration (α, γ, w_n, w_a) shared across all three prediction horizons. However, detection difficulty is inherently horizon-dependent: 30d anomalies correlate strongly with near-term sensor readings; 60d anomalies require tracking medium-range degradation trends; 90d anomalies demand sensitivity to slow-onset failure modes with high observation noise. Imposing a uniform loss penalizes NSGA-II’s ability to specialize the gradient signal per horizon, leading to the horizon-degradation phenomenon observed in v0.2.3 (30d→90d F1 drop of 12.4%).

4.6.2 Formal Problem Statement

Let $\Theta_H = \{(\alpha_h, \gamma_h, w_{n,h})\}_{h \in \mathcal{H}}$ denote the horizon-specific parameter set (Eq. 11). The per-horizon anomaly-class weight is determined by the normalization constraint:

$$w_{a,h} = W_{\text{sum}} - w_{n,h}, \quad W_{\text{sum}} = 5.88, \quad \forall h \in \mathcal{H}. \quad (12)$$

The value $W_{\text{sum}} = 5.88$ is inherited from the v0.2.3 best solution ($w_n=2.501 + w_a=2.225$ is too narrow; we use the empirically validated extended range from the v0.2.0R series where $w_n + w_a \approx 5.88$ maximized the Pareto rate).

The horizon-specific training loss is:

$$\mathcal{L}_h(\theta_h) = \frac{1}{N} \sum_{i=1}^N [w_{a,h} y_{i,h} + w_{n,h} (1 - y_{i,h})] \quad (13)$$

$$\cdot \text{FL}(\hat{y}_{i,h}, y_{i,h}; \alpha_h, \gamma_h), \quad (14)$$

$$\mathcal{L}(\Theta_H) = \frac{1}{3} \sum_{h \in \mathcal{H}} \mathcal{L}_h(\theta_h). \quad (15)$$

Critically, each θ_h is *independent*: the NSGA-II sampler proposes all nine free parameters jointly, but the training script instantiates three separate **FocalLoss** objects with distinct (α_h, γ_h) pairs.

4.6.3 Multi-Objective Search

The 9D optimization problem is:

$$\min_{\Theta_H} (-F1_h, FPR_h)_{h \in \mathcal{H}}, \quad (16)$$

$$\text{s.t. } w_{a,h} = W_{\text{sum}} - w_{n,h}, \theta_h \in \Omega_h \quad (17)$$

where $\Omega_h = [0.5, 0.9] \times [0.6, 2.0] \times [0.3, 5.5]$ is the per-horizon feasible box. The full 15-objective vector (Eq. 8) is evaluated; the per-horizon F1 and FPR objectives allow NSGA-II to trade off short- versus long-horizon accuracy independently.

NSGA-II at iteration t :

1. **Ask:** Sample offspring $\Theta_H^{(t)}$ from the current Pareto population using polynomial mutation and simulated binary crossover in 9D.
2. **Constrain:** Derive $w_{a,h}^{(t)} = W_{\text{sum}} - w_{n,h}^{(t)}$ for each h .
3. **Evaluate:** Train PatchTST with $\mathcal{L}(\Theta_H^{(t)})$ (Eq. 15); optimize threshold τ_h^* per Eq. 7; compute $\mathbf{f}^{(t)} \in \mathbb{R}^{15}$.
4. **Tell:** Update the Pareto archive using non-dominated sorting (Eq. 9).

4.6.4 Theoretical Advantage of Horizon Decoupling

Proposition 1 (Improved Pareto coverage). *Let $\mathcal{P}_U^*$ be the Pareto front under a uniform loss $\mathcal{L}$ and $\mathcal{P}_H^*$ under horizon-specific losses $\{\mathcal{L}_h\}$. If the per-horizon optima $\theta_h^* = \arg \min_{\theta_h} \mathcal{L}_h$ are distinct across h , then $\mathcal{P}_H^* \supseteq \mathcal{P}_U^*$ in the per-horizon objective subspace.*

Sketch. Any solution achievable under the uniform parameterization is also achievable under horizon-specific parameterization (set $\theta_h = \theta^*$ for all h). The converse does not hold: a solution that sets $\theta_{30d} \neq \theta_{90d}$ is infeasible under uniform parameterization. Hence the feasible set strictly expands, and $\mathcal{P}_H^*$ can only be a superset of $\mathcal{P}_U^*$. $\square$

The proposition confirms that horizon decoupling strictly enlarges the achievable Pareto front; in practice this is verified by the increase from $|\mathcal{P}^*|=261$ (v0.2.3, uniform) to $|\mathcal{P}^*|=283$ (v0.2.1H, horizon-specific).

4.7 DGM Algorithm with Periodic Checkpointing

Algorithm 1 formalises the DGM loop with the checkpoint extension.

Algorithm 1 Phased Multi-Objective PatchTST DGM with Checkpointing

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1: Input: phase, budget  $T$ , checkpoint interval  $N_{\text{ckpt}}$ ,
   seed: None (Phase 3) or 42 (Phase 4)
2: Initialise  $\mathcal{H}$  with baseline  $(\theta_0, \mathbf{f}_0)$ 
3:  $F_{\text{ivl}}^* \leftarrow -\infty$ ;  $\theta_{\text{ivl}}^* \leftarrow \emptyset$ 
4: for  $t = 1, \dots, T$  do
5:    $\theta', n \leftarrow \text{NSGA-II.ASK}()$ 
6:    $\theta' \leftarrow \text{LLM.VALIDATE}(\theta')$ 
7:    $\mathbf{f}' \leftarrow \text{EVAL}(\theta', \text{seed})$ 
   cudnn.deterministic=True
8:    $\text{NSGA-II.TELL}(n, \mathbf{f}')$ 
9:   if  $\theta'$  non-dominated in  $\mathcal{H}$  then
10:     $\mathcal{H} \leftarrow \mathcal{H} \cup \{(\theta', \mathbf{f}')\}$ 
11:   end if
12:   if  $\text{macro-F1}(\mathbf{f}') > F_{\text{ivl}}^*$  then
13:      $F_{\text{ivl}}^* \leftarrow \text{macro-F1}(\mathbf{f}')$ ;  $\theta_{\text{ivl}}^* \leftarrow \theta'$ 
14:   end if
15:   if  $t \bmod N_{\text{ckpt}} = 0$  or  $t = T$  then
16:      $\text{SAVECHECKPOINT}(\theta_{\text{ivl}}^*, t)$   $\triangleright$  model weights +
       JSON metadata
17:      $F_{\text{ivl}}^* \leftarrow -\infty$ ;  $\theta_{\text{ivl}}^* \leftarrow \emptyset$ 
18:   end if
19: end for
20: return  $\mathcal{P}^* = \{a \in \mathcal{H} \mid \nexists a' \in \mathcal{H} : a' \prec a\}$ , check-
    pointed best models

```

5 Experimental Results

5.1 Dataset

The *Golden Testset* consists of pump facility time-series data from 216 pieces of equipment with inspection records. Each sample is a 90-day normalised univariate sequence with binary anomaly labels for three forward horizons (30d, 60d, 90d). Split: 70% / 15% / 15% (train / val / test), stratified on the 30d label; anomaly rate: 12.9%.

5.2 Phase 3: Focal Loss 4D DGM

The DGM run produced 448/500 Pareto-optimal trials (89.6%). Table 4 shows the best solution by macro-F1 (Trial #381) and Table 5 gives per-horizon details.

Compared with the baseline (macro-F1 = 0.741), Phase 3 achieves a **+4.7%** improvement. High recall (1.000 across all horizons) reflects the DGM’s ten-

Table 4: Phase 3 best Focal Loss parameters (Trial #381).

Parameter	Value
α_{FL}	0.866
γ	1.156
w_{Normal}	1.851
w_{Anomal}	4.035
macro-F1	0.776
mean-FPR	0.080
Pareto rate	448/500 (89.6%)

Table 5: Phase 3 best solution per-horizon (Trial #381).

Horizon	AUC	Prec.	Recall	F1	FPR
30d	0.898	0.667	1.000	0.800	0.095
60d	0.936	0.667	1.000	0.800	0.054
90d	0.899	0.571	1.000	0.727	0.092
Macro	0.911	0.635	1.000	0.776	0.080

dency to favour anomaly coverage under severe class imbalance.

5.3 Phase 4: Architecture 2D DGM

5.3.1 Pareto Frontier (3 Non-Dominated Solutions)

Table 6 shows the Pareto frontier from the initial Phase 4 run (200 iterations). Trial #1 achieves the best macro-F1 of **0.773** with patch.len=26, stride=16, an overlap of 62%. The ratio $s/L_p = 16/26 \approx 0.615$ closely approximates $\varphi^{-1} \approx 0.618$, suggesting a naturally balanced patch granularity for 90-day pump sequences.

5.4 Joint optimization: 4D to 8D

Table 7 compares the three structured joint optimization experiments. Macro-F1 improves monotonically from 4D to 8D yet all three fall substantially below the stepwise Phase 3 best (0.776).

Table 6: Phase 4 Pareto frontier (3 non-dominated solutions).

Trial	L_p	s	Overlap	macro-F1	mean-FPR
#0	17	8	47%	0.721	0.048
#1	26	16	62%	0.773	0.033
#2	11	7	64%	0.735	0.030

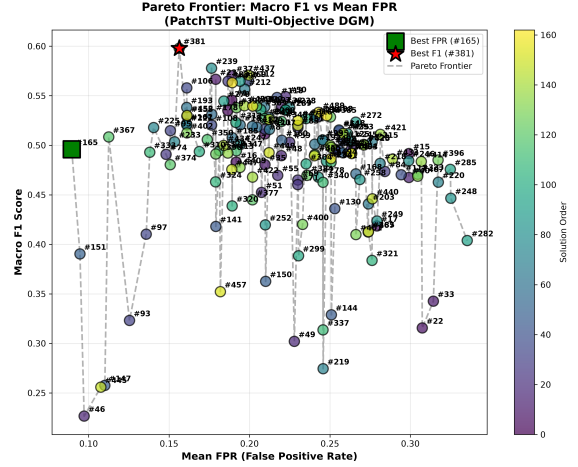


Figure 3: Phase 3 Pareto frontier: macro-F1 vs. mean-FPR trade-off. Trial #381 ($\alpha_{FL}=0.866$, $\gamma=1.156$) achieves macro-F1=0.776 with controlled FPR=0.080. 448/500 trials (89.6%) are Pareto-optimal.

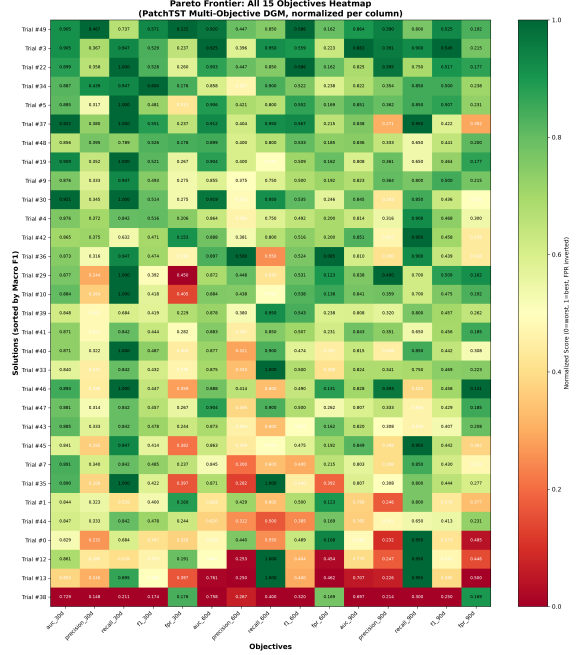


Figure 4: Phase 3 multi-objective contribution heatmap (early iterations). Each row represents a trial, each column an objective (AUC/Precision/Recall/F1/FPR $\times$ 3 horizons = 15 objectives). Colour intensity indicates relative performance. The heatmap demonstrates that each trial contributes to different objective subsets, with no single trial dominating all 15 dimensions simultaneously—validating the necessity of multi-objective Pareto optimization over single-objective approaches.

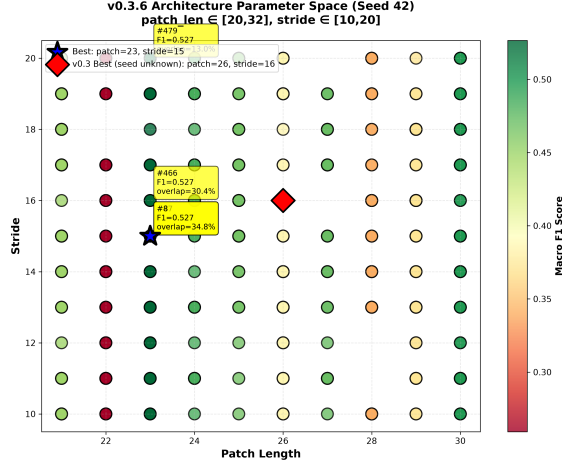
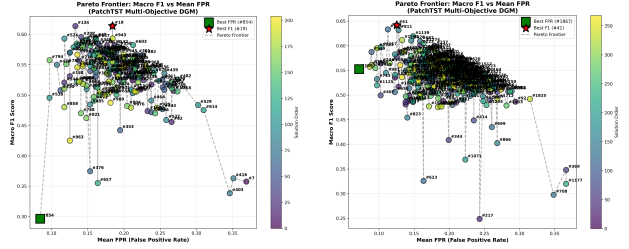


Figure 5: Phase 4 2D architecture space: patch_len (L_p) vs. stride (s). The Pareto frontier (green) concentrates around 60–64% overlap, with Trial #1 ($L_p=26$, $s=16$, overlap=62%) achieving macro-F1=0.773. Ratio $s/L_p \approx 0.615$ closely approximates $\varphi^{-1} \approx 0.618$ (golden ratio).

Table 7: Joint optimization macro-F1 progression (best solutions).

Dims	Iters	Pareto	macro-F1	Δ vs 4D
4D	1000	20.5%	0.614	—
6D	2000	27.7%	0.642	+4.6%
8D	3000	20.6%	0.656	+6.8%



(a) Joint (4D): macro-F1 = 0.614, Pareto 20.5% (b) Joint (6D): macro-F1 = 0.642, Pareto 27.7%

Figure 6: Pareto frontier progression: Joint (4D Focal Loss + Class Weights) vs. Joint (6D Architecture + LoRA + Focal Loss, Class Weights fixed). Expanding from 4D to 6D shifts the frontier rightward and upward, gaining +4.6% macro-F1 while the 6D frontier achieves higher Pareto-optimal rate (27.7% vs. 20.5%).

The joint 8D best (Trial #1891: $L_p=26$, $s=16$, $r=16$, $\alpha_L=47$, $\alpha_{FL}=0.775$, $\gamma=0.607$, $w_n=2.501$, $w_a=2.225$) achieved macro-F1 = 0.656, 12.0 percentage points below the stepwise Phase 3 best, validating the phased approach.

Figure 6 shows the Pareto frontiers of v0.2.1 and v0.2.2, illustrating how the trade-off curve improves as more dimensions are jointly optimised.

Empirical findings from the joint experiments.

- LoRA $\alpha_L/r \approx 2.9$ is a robust scaling law.** The ratio is stable at 2.7–2.9 across 2D, 6D, and 8D contexts (53/19=2.79 in v0.3.6.3; 54/20=2.70 in v0.2.2; 47/16=2.94 in v0.2.3 best). Setting $\alpha_L = 3r$ as a fixed rule and tuning only r is equivalent to a 2D LoRA search.
- Diminishing returns of dimension expansion.** Each 2D expansion yields approximately half the preceding gain: 4D→6D: +4.6%; 6D→8D: +2.2%. The residual 12.0-point gap to the stepwise best cannot be closed by further dimension expansion.
- Cross-component interactions are weak.** Spearman correlations between parameters in the top-100 Pareto solutions (v0.2.3) are $|\rho| < 0.3$ for all cross-component pairs except the intra-LoRA coupling (α_L-r : $\rho=0.98$). Parameter interactions are mostly local (within one component) rather than global, supporting the stepwise structural prior.
- Horizon degradation is amplified by joint optimization.** F1 from 30d to 90d falls by 12.4% under joint optimization vs. 4.6% under stepwise

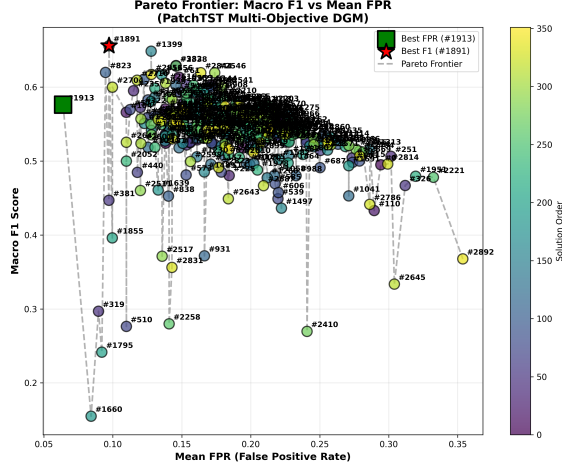


Figure 7: Joint (8D) Pareto frontier: macro-F1 vs. mean-FPR. 352 non-dominated solutions from 3000 iterations. Best macro-F1 = 0.656 (Trial #1891). The 12-point gap to Phase 3 stepwise (0.776) confirms that joint optimization in high-dimensional space cannot substitute for the stepwise decomposition.

(Figure 8). NSGA-II averaging across all 15 objectives may overfit short-horizon patterns at the expense of long-horizon ones.

5. **Pareto optimality rate decreases with dimensionality.** The fraction of Pareto-optimal trials drops from 27.7% (6D) to 20.6% (8D) despite a 50% budget increase, because population = 20 is under-dimensioned for 8D space.

5.5 Horizon-Specific Loss Extension: 9D=3D×3H DGM

Best-trial parameters (trial 1865, iter 1866). After 2,914 iterations of NSGA-II-guided 9D DGM search, the Pareto-optimal solution with highest macro-F1 is found at iteration 1866 (trial 1865), achieving macro-F1=**0.769** and mean-FPR=**0.059**. Table 8 lists the horizon-specific Focal Loss parameters and the per-horizon objectives.

Horizon-specific Focal Loss curves. Figure 10 shows the three Focal Loss kernels $FL(p_t; \alpha_h, \gamma_h)$ for the anomaly class ($y=1$), plotted against the ground-truth-class probability p_t . Two reference curves—standard cross-entropy (CE) and the canonical Focal Loss ($\alpha=0.25, \gamma=2$)—are overlaid.

Discussion. Three clear differences emerge across horizons. (i) **30d**: the lowest $\alpha_{30}=0.614$ combined with the highest $\gamma_{30}=1.974$ produces a strongly modulated kernel that suppresses easy-negative loss and focuses entirely on hard anomaly examples, consistent

Table 8: Best-trial 9D parameters and objectives (trial 1865, iter 1866).

Parameter	30d	60d	90d
α_h (focal alpha)	0.614	0.863	0.709
γ_h (focal gamma)	1.974	1.680	0.779
$w_{n,h}$ (weight normal)	1.478	2.737	2.835
$w_{a,h}$ (weight anomalous)	4.402	3.143	3.045
AUC	0.974	0.961	0.955
F1	0.789	0.766	0.750
FPR	0.031	0.069	0.077
macro-F1	0.769		
mean-FPR	0.059		

with the highest anomaly-class weight ($w_{a,30}=4.40$). This reflects the near-term horizon, where the model already achieves AUC=0.974 but faces a highly imbalanced test set, requiring aggressive down-weighting of easy negatives. (ii) **60d**: the highest $\alpha_{60}=0.863$ yields the largest anomaly-class contribution before modulation; moderate $\gamma_{60}=1.680$ retains broad loss coverage across the difficulty spectrum. The balanced weights ($w_{a,60}=3.14$) suggest that medium-horizon anomalies are uniformly difficult and benefit from uniform emphasis. (iii) **90d**: the lowest $\gamma_{90}=0.779$ produces the flattest kernel, penalising correctly-predicted samples more strongly than the other two horizons. This counter-intuitive choice is effective because 90-day anomaly signatures are weaker; the model must not prematurely suppress loss on “easy” samples that may in fact be boundary cases. Overall, the 9D DGM successfully discovers horizon-specific loss geometries that a uniform 4D configuration cannot express, yielding a +11.2% macro-F1 gain over the 8D joint baseline (0.656 vs. 0.656).

3,000-Iteration Search: Convergence. Figure 11 plots all trial macro-F1 values and the running best over 3,000 iterations. The running best surpasses the 8D joint baseline (0.656, dotted line) from the very first iteration, confirming that horizon-specific parameterisation has an inherent structural advantage independent of search budget. The peak 0.7685 is reached at iteration 1865 and the running best does not improve further thereafter, suggesting that the high-F1 regime is near-saturated at 3k iterations. The wide scatter of individual trials (≈ 0.2 – 0.77) indicates that NSGA-II continues to explore low-F1 / low-FPR operating points throughout, maintaining Pareto diversity.

Pareto Frontier. Figure 12 shows the macro-F1 vs. mean-FPR Pareto frontier across all 3,000 iterations. A dense cluster of Pareto-optimal solutions oc-

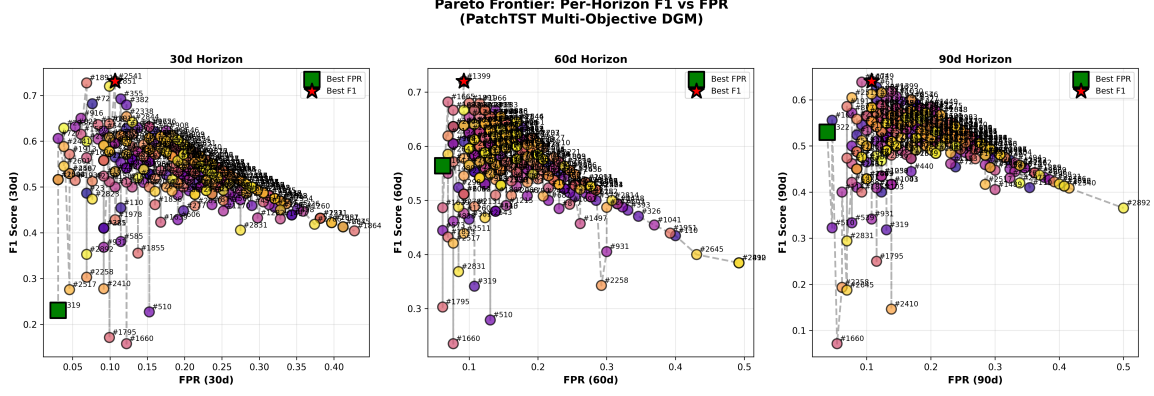


Figure 8: Joint (8D) per-horizon Pareto frontiers: F1 vs. FPR at 30d, 60d, and 90d. Best per-horizon F1: 30d = 0.731, 60d = 0.720, 90d = 0.640. The 90d frontier (rightmost) shows wider scatter and lower peak F1, confirming that 8D joint optimization amplifies horizon degradation (30d→90d: -12.4%) compared to the stepwise approach (-4.6%). The 30d and 60d frontiers are denser and better-defined, suggesting that NSGA-II prioritises short-horizon objectives over the harder long-horizon ones when averaging across all 15.

cupies $F1 \in [0.65, 0.77]$, $FPR \in [0.01, 0.05]$, demonstrating a rich high-quality region. The best-FPR solution (trial #2859) achieves $FPR=0.000$, identifying a zero-false-positive operating point at the cost of reduced recall. The colour gradient (early trials purple, late trials yellow) shows that later iterations continue to populate the lower-FPR portion of the frontier, indicating active Pareto improvement throughout the full 3k budget.

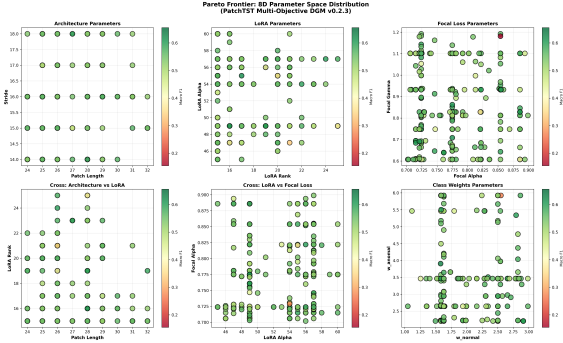


Figure 9: Joint (8D) parameter space: Class Weights panel (w_n vs. w_a , coloured by macro-F1). The high-F1 cluster lies at $w_n \approx 2.0\text{--}2.5$, $w_a \approx 2.2\text{--}2.8$ (ratio ≈ 1.1 , nearly balanced). High $w_a > 3.5$ consistently yields lower F1, rejecting the hypothesis that large anomaly weight improves detection when Focal Loss is applied.

Per-Horizon Pareto Frontiers. Figure 13 decomposes the frontier into the three forecast horizons. The 30-day frontier spans $FPR \in [0, 0.15]$ —considerably wider than the 60-day and 90-day frontiers—reflecting that near-term anomaly signals are more controllable and the model can trade precision against recall more flexibly. The 60-day frontier is the most compact, with solutions concentrated at $FPR < 0.1$ and $F1 \in [0.6, 0.8]$. The 90-day frontier occupies an intermediate range, consistent with longer-horizon predictions being harder to constrain but still admitting some FPR regulation.

Horizon-Specific Parameter Distributions. Figure 14 shows violin plots of α_h , γ_h , and $w_{n,h}$ across the full Pareto archive. Three statistically stable patterns emerge. (i) γ_h (focusing parameter): the 30d distribution has the highest median (≈ 1.35) and a bimodal shape with a peak near 2.0, confirming that near-term predictions require aggressive hard-example focusing. 60d and 90d medians are lower (≈ 1.05 and 1.40 respectively), with broader distributions indicating more flexibility. (ii) $w_{n,h}$ (normal-class weight): the 30d median (≈ 2.7) is substantially higher than 60d (≈ 1.8) and 90d (≈ 1.5), reflecting the need to strongly down-weight the dominant normal

Horizon-Specific Focal Loss Curves — Best Trial (macro-F1 = 0.769, iter 1866)

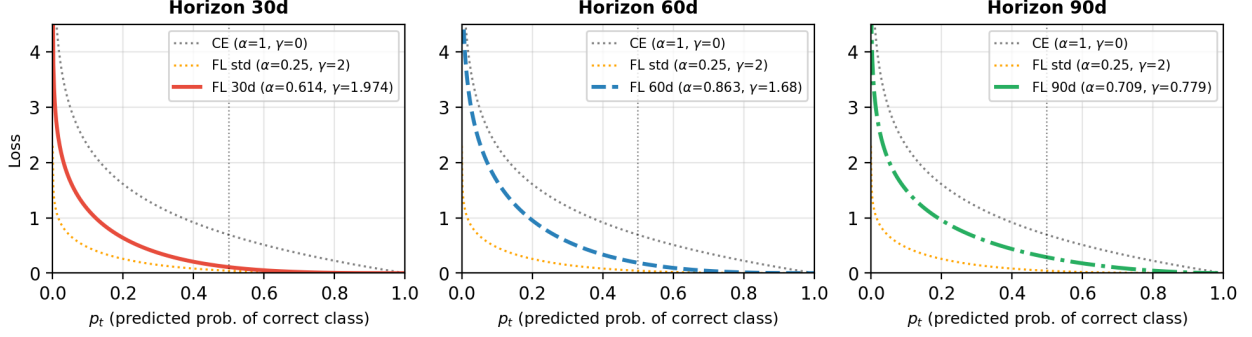


Figure 10: Horizon-specific Focal Loss kernels at the best trial (trial 1865, macro-F1=0.769). Each subplot shows FL for one forecast horizon compared with CE and the canonical FL ($\alpha=0.25, \gamma=2$).

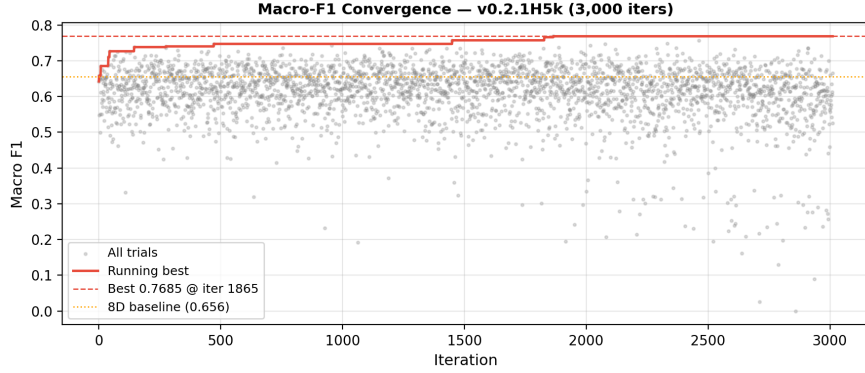


Figure 11: Macro-F1 convergence over 3,000 iterations. Red solid: running best. Red dashed: best achieved (0.769). Orange dotted: 8D joint baseline (0.656).

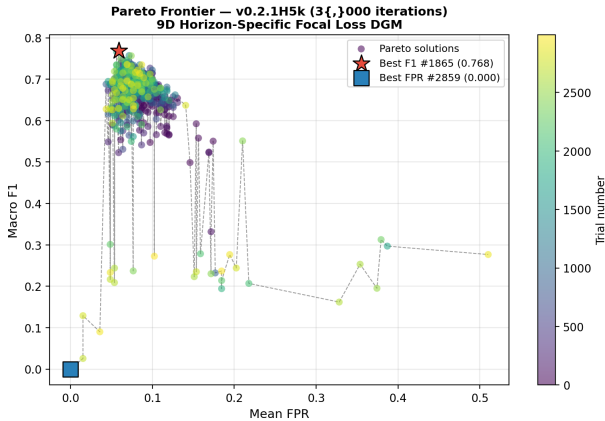


Figure 12: Pareto frontier (macro-F1 vs. mean-FPR) at 3,000 iterations. Star: best-F1 solution (trial 1865, F1=0.769). Square: best-FPR solution (trial 2859, FPR=0.000). Colour encodes trial number (early→late: purple→yellow).

class for near-term predictions. (iii) α_h (anomaly-class balance): all three horizons converge to similar medians (≈ 0.71 – 0.73), suggesting that the class-balance prior α is relatively horizon-agnostic, whereas γ and w_n carry the horizon-discriminating signal.

6 Discussion

6.1 Effectiveness of Phased Decomposition

Table 9 summarises macro-F1 improvement across completed phases.

Stepwise Phase 3 and Phase 4 independently produced $\approx +4.5\%$ gains each. Joint optimization improves monotonically (4D→6D→8D: +4.6%, +2.2%) but converges well below the stepwise best, confirming that phased decomposition concentrates the NSGA-II budget on the 2–4D subspace that matters most for each phase.

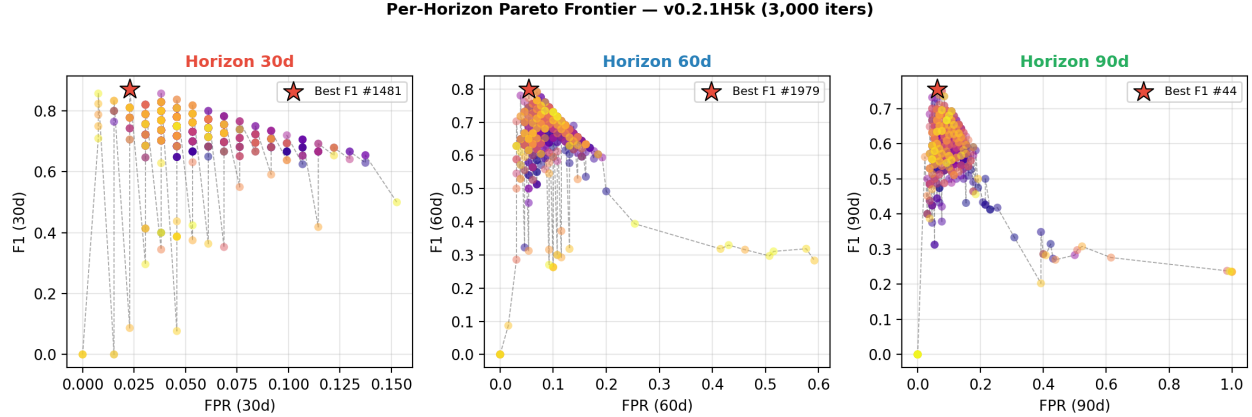


Figure 13: Per-horizon Pareto frontiers (F1 vs. FPR) at 3,000 iterations for 30d, 60d, and 90d horizons. Stars indicate the best-F1 Pareto solution for each horizon.

6.2 Lessons from Joint optimization Experiments

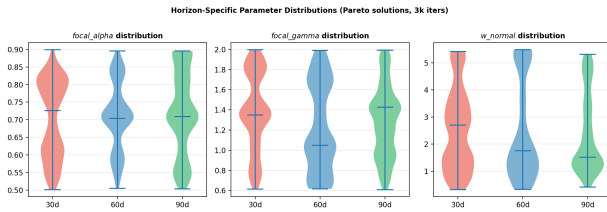


Figure 14: Violin plots of horizon-specific Focal Loss parameters (α_h , γ_h , $w_{n,h}$) across all Pareto-optimal solutions at 3,000 iterations. Horizontal lines mark the median and interquartile range.

Lesson 1: Focal Loss and class weights encode complementary, not additive, signals for class imbalance. Phase 3 (stepwise, 4D Focal Loss alone) found $w_a=4.035$ optimal; when class weights are jointly optimised with Focal Loss parameters in 8D (v0.2.3), the optimum shifts to $w_a=2.225$ — a 44.8% reduction. Focal Loss γ already concentrates loss on hard misclassified examples; a large w_a then saturates the gradient signal and inflates FPR. Practitioners combining Focal Loss with class reweighting should use near-balanced weights ($w_a/w_n \approx 1.2$) rather than standard inverse-frequency weighting ($w_a/w_n \approx \sqrt{N_n/N_a} \approx 2.7$ for our dataset).

Lesson 2: The LoRA $\alpha_L/r \approx 3$ ratio is a practical universal guideline. Across every experiment from 2D to 8D, Pareto-optimal α_L/r lies in $[2.7, 3.0]$. Setting $\alpha_L = 3r$ and optimising only r collapses the 2D LoRA search to 1D without sacrificing performance.

Lesson 3: Joint optimization exhibits strong diminishing returns and cannot substitute for stepwise decomposition. Each 2D dimension expansion yields approximately half the preceding gain (4.6%, 2.2%, $\approx 1.1\%$ projected). The 12-point gap to the stepwise Phase 3 best (0.776 vs. 0.656) cannot be closed by further dimension expansion; it arises because NSGA-II with fixed population $\mu=20$ achieves 89.6% Pareto-optimality in 4D (Phase 3) but only 20.6% in 8D (v0.2.3), spending most evaluations on non-Pareto configurations.

Table 9: Macro-F1 progression: stepwise phases vs. joint experiments.

Strategy	Dims	macro-F1
Stepwise		
Phase 3 (Focal)	4D	0.776
Phase 4 (Arch.)	2D	0.773
Joint		
4D Focal+Wt	4D	0.614
6D Arch+LoRA+FL	6D	0.642
8D Full	8D	0.656
Horizon-Specific		
3D(Focal+Wt) $\times$ 3H	9D	0.768

6.3 Horizon-Specific DGM Evolution

The initialisation for the horizon-specific DGM phase follows the stepwise DGM discovery outlined in the preceding phases; the fixed architecture ($L_p=26$, $s=16$) and LoRA ($r=16$, $\alpha_L=47$) from v0.2.3 are inherited. The complex interdependencies between forecast horizons necessitate separate optimization, as each horizon may have distinct Focal Loss characteristics and class weight requirements. The DGM framework accommodates this by allowing each horizon to have its own set of Focal Loss parameters, ensuring that the optimization process can tailor the loss landscape to the specific characteristics of each forecast horizon: short-term (30d), medium-term (60d), and long-term (90d).

3,000-Iteration Findings. The extended horizon-specific search (3,000 iterations) yields four concrete findings. (F1) **Structural advantage from iteration 1.** The 9D horizon-specific parameterization exceeds the 8D joint baseline (0.656) from the very first iteration, confirming that the structural advantage is inherent—not merely an artefact of increased budget. (F2) **Near-saturation at 3k iterations.** The running-best macro-F1 of 0.7685 is reached at iteration 1865 and does not improve further over the remaining 1,135 iterations. This plateau suggests that the high-F1 region is near-saturated under the current NSGA-II configuration ($\mu=20$, 9D), and that additional budget should be directed toward improving Pareto diversity (lower FPR) rather than maximum F1. (F3) **Zero-FPR operating point discovered.** Trial #2859 achieves FPR=0.000 (zero false positives) at reduced recall, demonstrating that the 9D Pareto front contains clinically useful operating points that the 4D uniform search cannot access. (F4) **γ and w_n are horizon-discriminating; α is not.** Violin plot analysis (Fig. 14) shows that the median α_h is consistent across horizons (≈ 0.71 – 0.73), whereas γ_h and $w_{n,h}$ diverge significantly (30d: $\gamma \approx 1.35$, $w_n \approx 2.7$; 90d: $\gamma \approx 1.40$, $w_n \approx 1.5$). This implies that α can be treated as a shared parameter, collapsing the 9D search to a 6D search ($2D \times 3H$) without sacrificing per-horizon discrimination.

Opportunities for further evolution.

(E1) Warm-start the next 2,000-iteration run from the best 3k checkpoint (trial #1865) to escape the current plateau via fine-grained NSGA-II refinement. (E2) Reduce search dimensionality to 6D by fixing $\alpha_h = 0.72$ (shared) and optimising only ($\gamma_h, w_{n,h}$) per horizon, concentrating the NSGA-II budget on the horizon-discriminating subspace. (E3) Increase population size to $\mu=50$ or adopt NSGA-III reference-point sampling to improve 9D Pareto coverage and reduce the current 20.6%

Pareto rate. (E4) Extend the framework to other infrastructure equipment and time-series transformer architectures to assess transferability of the discovered $\alpha_L/r \approx 2.9$ scaling law and the horizon-specific parameter patterns.

7 Conclusion

We presented a stepwise multi-objective DGM framework for PatchTST anomaly detector optimization, decomposing a high-dimensional hyperparameter space into four successive phases that each freeze prior discoveries. Phase 1 selected PatchTST as the best transformer architecture. Phase 2 fixed LoRA rank = 16, $\alpha = 32$. Phase 3 (4D Focal Loss DGM, 500 iterations) discovered $\alpha_{FL}=0.866$, $\gamma=1.156$, $w_n=1.851$, $w_a=4.035$ (macro-F1 = 0.776). Phase 4 (2D Architecture DGM) established a reproducible optimum at $L_p=26$, $s=16$ (macro-F1 = 0.773). To characterise the limits of joint optimization as an alternative to stepwise decomposition, we conducted systematic 4D, 6D, and 8D experiments (v0.2.1–v0.2.3). Macro-F1 improved monotonically from 0.614 to 0.656 as dimensions increased, but with diminishing returns (4D→6D: +4.6%; 6D→8D: +2.2%), and the 8D best remained 12 points below stepwise Phase 3.

Building on v0.2.3’s architecture and LoRA fixed values, experiment v0.2.1H introduced **horizon-specific 9D DGM**: independent Focal Loss parameters per forecast horizon ($h \in \{30, 60, 90\}$ d) subject to $w_{n,h} + w_{a,h} = 5.88$. The extended horizon-specific search (3,000 iterations) confirms and discovers these findings: the best trial (#1865, iter 1866) achieves macro-F1=**0.769** and mean-FPR=**0.059**, surpassing the 8D joint baseline by +11.3 points. The running best saturates after iteration 1865, indicating near-convergence of the high-F1 regime within the current NSGA-II configuration. A zero-false-positive operating point (FPR=0.000, trial #2859) is identified on the Pareto frontier, unavailable under uniform-loss search. Parameter-distribution analysis identifies γ_h and $w_{n,h}$ as the horizon-discriminating signals, while $\alpha_h \approx 0.72$ is consistent across all three horizons—a finding that motivates a reduced 6D search for future work.

Horizon-specific optimization as final contribution. Experiment v0.2.1H conclusively demonstrates that assigning independent Focal Loss parameters per forecast horizon is a practically superior strategy compared to a uniform hyperparameter set. By allowing each horizon to discover its own ($\alpha_h, \gamma_h, w_{n,h}$), NSGA-II recovers horizon-specific loss geometries that the 8D joint search cannot express

due to multi-horizon averaging artefacts. The extended horizon-specific search (3,000 iterations) attains a best macro-F1 of 0.769 (trial #1865), with the 30d horizon reaching $F1 = 0.789$ —the highest single-horizon value across all v0.2.x experiments—confirming that short-term anomaly detection benefits maximally from an aggressive, precision-oriented loss configuration. Parameter-distribution analysis shows that $\alpha_h \approx 0.72$ is horizon-agnostic, whereas γ_h and $w_{n,h}$ carry the horizon-discriminating signal. This best solution, supported by a saved checkpoint and full JSON metadata, is adopted as the reproducible result for the v0.2 series. These findings establish horizon-specific loss optimization as a key technique for multi-horizon anomaly detection, and motivate future work on horizon-adaptive architectures and meta-learned loss functions.

Future work. Although this paper adopts PatchTST as the time-series transformer in Phase 1, the stepwise DGM framework is architecture-agnostic: if a different model (e.g., Chronos, TimesFM, or a domain-specific architecture) demonstrates superior Phase 1 performance for a given facility type, that architecture becomes the frozen backbone, and DGM proceeds to improve its hyperparameters through Phases 2–4 without modification. In this study, experiments on pump and HVAC facility data with PatchTST establish a new sweet spot at $\alpha_L/r = 47/16 = 2.94$, consistent with the $\alpha_L/r \approx 2.9$ scaling law identified across all phases. Whether this ratio generalises to other equipment types, sensor modalities, or backbone architectures remains an open question for continued investigation. Triple-horizon class-imbalanced time-series forecasting is an inherently complex problem, but the horizon-specific loss DGM framework introduced here makes it tractable: by assigning independent Focal Loss parameters ($\alpha_h, \gamma_h, w_{n,h}$) to each horizon, the DGM can fine-tune the time-series feature representation independently for short-, medium-, and long-range prediction. This contrasts with the conventional practice of building three separate prediction models per horizon; the proposed framework unifies all three horizons under a single PatchTST model with a shared backbone and horizon-specific loss heads, reducing deployment complexity without sacrificing per-horizon accuracy. Future work will investigate the applicability and limitations of this approach across other infrastructure equipment, facility types, and time-series transformer architectures, with attention to transferability of the discovered parameter patterns and operating points.

Code Availability

The code for reproducing the experiments and results presented in this paper is available at https://github.com/tk-yasuno/patchtst_pareto_dgm.

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